ELEMENTARY ETALE NEIGHBOURHOODS OF A DECENT POINT FORM A DIRECTED SYSTEM WITH A FINAL GROUPOID QUOTIENT

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Editorial note. No omission markers in-span; the groupoid construction is delegated to an uncarded chapter lemma and declared external.

Allowed standard prerequisites (editorial).

- Quotient groupoid of an etale neighbourhood. lemma-etale-locally-quasi-finite-groupoid (same chapter, NOT carded, declared external). Gives the groupoid (X x_X x, R) with finite fibres over the point.
- Decency base of the neighbourhood topology. Spaces, lemmas base-change-descent (cross-chapter, named). Used in the directedness argument.

Source reference. Original reference 04AZ: focused-seeds/geometry/S9-decent-elementary-etale-neighbourhoods/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BBP: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2641–2641. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0EMX: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2621–2621. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03FV: focused-seeds/geometry/S9-decent-elementary-etale-neighbourhoods/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Let x be a point of X. The category of elementary étale neighborhoods of (X, x) is cofiltered (see Categories, Definition 04AZ).*

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Proof. The category is nonempty by Lemma 0BBP. Suppose that we have two elementary étale neighbourhoods $(U_i, u_i) \to (X, x)$. Then consider $U = U_1 \times_X U_2$. Since $\mathrm{Spec}(\kappa(u_i)) \to X$, $i = 1, 2$ are both monomorphisms in the class of x (Lemma 0EMX) , we see that

$$u = \mathrm{Spec}(\kappa(u_1)) \times_X \mathrm{Spec}(\kappa(u_2))$$

is the spectrum of a field $\kappa(u)$ such that the induced maps $\kappa(u_i) \to \kappa(u)$ are isomorphisms. Then $u \to U$ is a point of U and we see that $(U, u) \to (X, x)$ is an elementary étale neighbourhood dominating (U_i, u_i). If $a, b : (U_1, u_1) \to (U_2, u_2)$ are two morphisms between our elementary étale neighbourhoods, then we consider the scheme

$$U = U_1 \times_{(a,b),(U_2 \times_X U_2),\Delta} U_2$$

Using Properties of Spaces, Lemma 03FV we see that $U \to X$ is étale. Moreover, in exactly the same manner as before we see that U has a point u such that $(U, u) \to (X, x)$ is an elementary étale neighbourhood. Finally, $U \to U_1$ equalizes a and b and the proof is finished. □